

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Pharm (Quality Assurance)

University Theory Examination March/April 2018

PHQA003 Pharmaceutical Analysis – I

Date: 28/03/2018, Wednesday

Time: 10.00 a.m to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A)** Describe different grades of reagents and types of glassware used in pharmaceutical analysis **05**
- B)** What is Calibration? Differentiate calibration and validation. **05**
- Q.2** Narrate the pharmacopoeial tests for Tablets and Capsules. **10**

OR

- Q.2** How will you calibrate UV/Visible Spectrophotometer? Describe. **10**
- Q.3** **Answer the following (Any three)** **15**
- (A)** Explain the role of physicochemical properties of drug on its biopharmaceutical behavior.
- (B)** How pH meter, analytical balance and melting point apparatus are calibrated?
- (C)** Describe strategies for development of UV method for quantitative analysis of drug substance.
- (D)** Differentiate following pairs:
- (1) Accuracy and Precision
- (2) LOD and LOQ

SECTION-II

- Q.4** What is analytical method validation? Why it is required? Write definition, methodology and acceptance criteria for following validation parameters. **10**
- (1) Linearity (2) Specificity
- Q.5** Write about general methods used to determine quality of raw material used in cosmetic preparations. **10**

OR

- Q.5** What are the legislations and standards for food products in India? Write general chemical and instrumental methods for analysis of food ingredients. **10**
- Q.6** **Write short note on the following (Any three)** **15**
- (A)** LIMS
- (B)** Application of Experimental design in analytical method validation
- (C)** Pharmacopoeial tests for Vitamins
- (D)** Calibration of HPLC